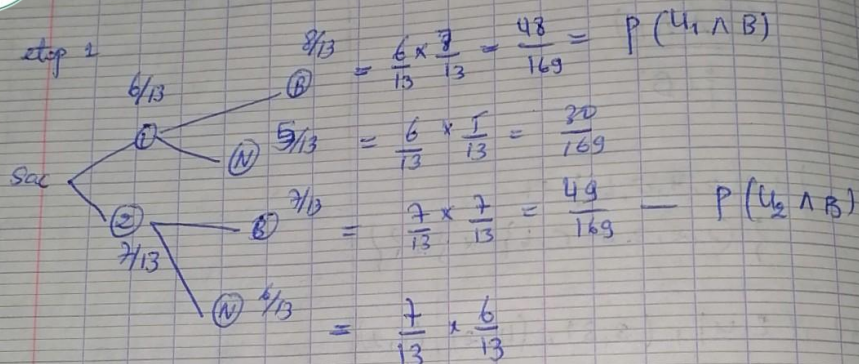
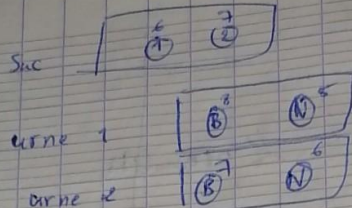




Ex 12

Ex 12 :



1) B = évé « la boule est blanche »

$$P(B) = P(U_1 \cap B) + P(U_2 \cap B)$$

$$= \frac{48}{169} + \frac{49}{169} = \frac{97}{169}$$

$$2) P_B(U_1) = \frac{P(U_1 \cap B)}{P(B)} = \frac{\frac{48}{169}}{\frac{97}{169}} = \frac{48}{97}$$



Exo 3

A) 1) $n = 90$, $p = 0,25$
 $X \sim B(90, 0,25)$
 $E(X) = np$
 $V(X) = np(1-p)$
 $p(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

2) $E(X) = 90 \times 0,25$

$V(X) = \frac{45}{2} \left(\frac{3}{4}\right) = 14,0625$

3) $p(X \geq 1) = 1 - p(X=0)$
 $= 1 - \left(\binom{90}{0} \times 0,25^0 \left(\frac{3}{4}\right)^{90} \right)$
 $p(X \geq 1) = 0,99$



B) $n = 25$, $p = 0,2$
 $X \sim B(25, 0,2)$

$p(X \geq 5) = 1 - (p(X=0) + p(X=1) + p(X=2) + p(X=3) + p(X=4))$

Exo 4.19 $N = 18$, $m = 7$, $n = 5$
 $X \sim J(18, 7, 5)$

$\begin{cases} p(X=k) \\ E(X) \\ V(X) \end{cases}$

2) $E(X) = \frac{nm}{N} = \frac{7 \times 5}{18}$

3) $p(X \leq 3) = p(X=0) + p(X=1) + p(X=2) + p(X=3)$



$p(X=0) = \binom{18}{0} \left(\frac{7}{18}\right)^0 \left(\frac{11}{18}\right)^{18}$